

The optimal trading rate is:

$$\alpha^* = \frac{X \cdot \sigma^2 \cdot P_0^2}{3 \cdot 250 \cdot ADV \cdot \mathcal{R}^{*2}} \quad (8.17)$$

Figure 8.2b illustrates this trading goal through strategy A2. Here, A2 is the strategy that minimizes cost for a risk exposure of \$0.40/share. It has an expected cost of \$0.20/share and corresponds to a trading rate of 25% and a POV rate of 20%.

3. Minimize Risk with Cost Constraint

A portfolio manager's preferred investment stock is LMK with an expected return of 10% and the next most attractive stock is RLK with an expected return of 9.5%. The manager determines that X shares can be purchased at a cost of 50 bp (0.50%). Purchasing any more shares of LMK will cause the cost to be greater than the incremental return of 50 bp and the manager would be better off investing some portion of the dollars in the second most attractive stock. Therefore, the manager decides to transact the X shares using a strategy that will minimize risk for a cost of 50 bp. This optimization is as follows:

$$\begin{aligned} & \text{Min } TR' \\ \text{s.t. } & \text{Cost}' = C^* \end{aligned} \quad (8.18)$$

Example: An investor looking to minimize timing risk for a specified level of market impact cost (not including price appreciation) will determine the optimal trading rate through minimizing the following equation:

$$\begin{aligned} \text{Min: } & \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{X}{ADV}} \cdot \alpha^{-1} \cdot P_0 \\ \text{s.t. } & b_1 \cdot I' \cdot \alpha + (1 - b_1) \cdot I' = C^* \end{aligned} \quad (8.19)$$

Solving for the optimal trading rate we get:

$$\alpha^* = \frac{C^* - (1 - b_1) \cdot I}{b_1 \cdot I} \quad (8.20)$$

Figure 8.2c illustrates the BES for an investor with a maximum cost of \$0.10/share. Here, strategy A3 has the lowest timing risk of \$0.73/share, and corresponds to a trading rate of 7% and a POV rate of 6.5%.

4. Balance Trade-off between Cost and Risk

The fourth criterion “Balance Trade-off between Cost and Risk” is used by investors with a certain level of risk aversion defined by the parameter λ . Risk adverse investors will set λ to be high to avoid market exposure and risk